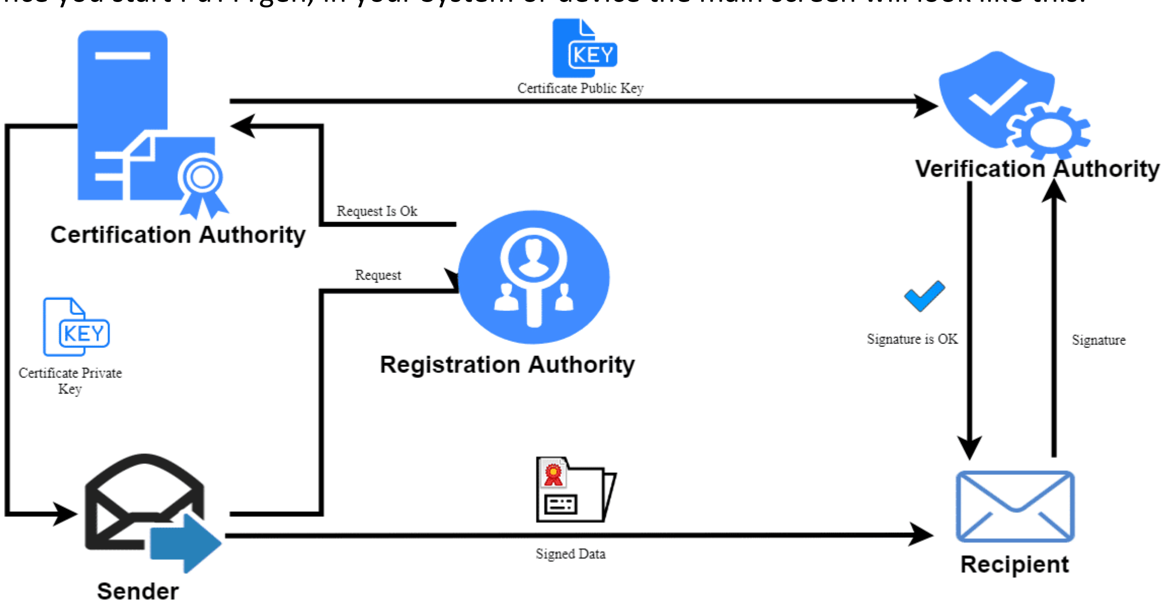
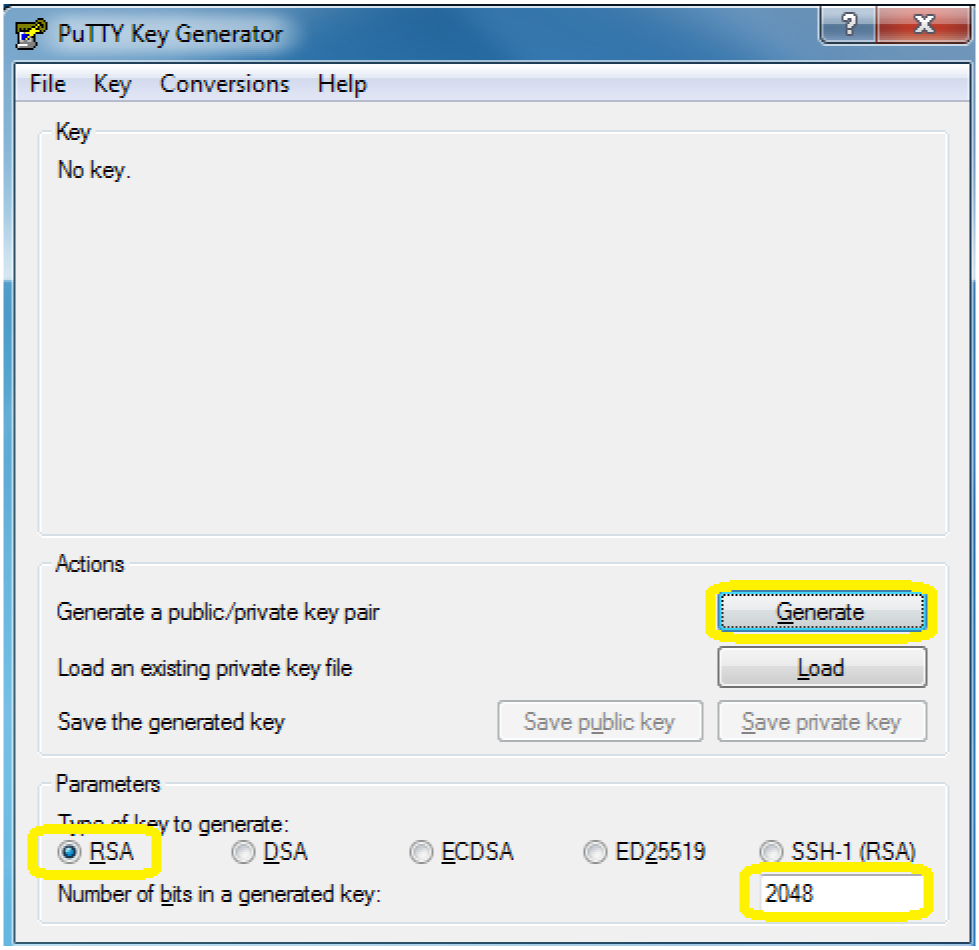
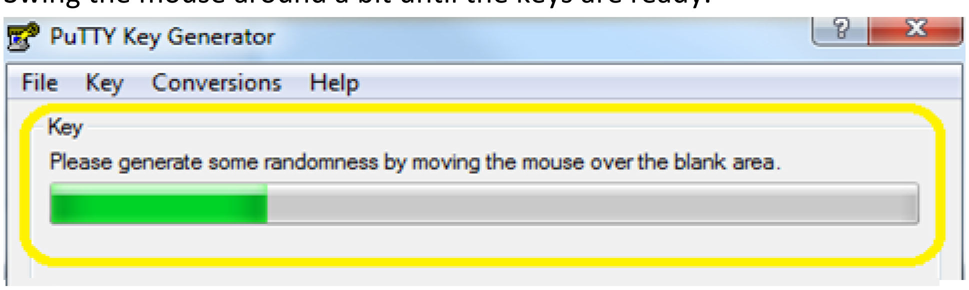
Public Key Infrastructure (PKI):

- The Public Key Infrastructure (PKI) is a set of hardware, software, people, policies.
- Procedures needed to create, manage, store, distribute, and revoke Digital Certificates.
- PKI enables users of a basically unsecure public network such as the Internet to securely.
- And privately exchange data through the use of a public & private cryptographic key pair.
- That Public Key Infrastructure is obtained and shared through a trusted authority.
- The core idea behind this system is to ensure that public key is used only by its owner.
- PKI binds each public key with identity of its owner with help of a digital SSL certificate.
- A Public Key Infrastructure implementation typically consists of following 4 elements:
- Certificate Authority (CA), which acts as “the root of trust”, CA issues digital certificates.
- Second element Registration Authority (RA), also known sometimes as a subordinate CA.
- Registration Authority (RA), validates registration of a digital certificate with a public key.
- Every time when a request for verification of any digital certificate is made, it goes to RA.
- The Registration Authority (RA), then decides whether to authenticate the request or not.
- Third element of PKI is a certificate database, which issues and revokes the certificates.
- And finally, a certificate store where issued certificates are stored with their private keys.
- Certificate store resides on local computer and stores issued certificates and private keys.
- PKI is an authentication method that uses key pair for authentication instead of password.
- The Public Key Infrastructure (PKI) two keys are generated: Public key and Private Key.
- Anyone that has public key is able to encrypt data that can be decrypted by the private key.
- Can share public key with anyone want, & they will be able to send encrypted messages.
- In the Public Key Infrastructure (PKI) The private key has to be protected not to be shared.
- Let us add the public key to a Cisco IOS Router and use it for SSH authentication method.
- Router will send encrypted messages that only can decrypt because we have private key.
- Once you start PuTTYgen, in your System or device the main screen will look like this.

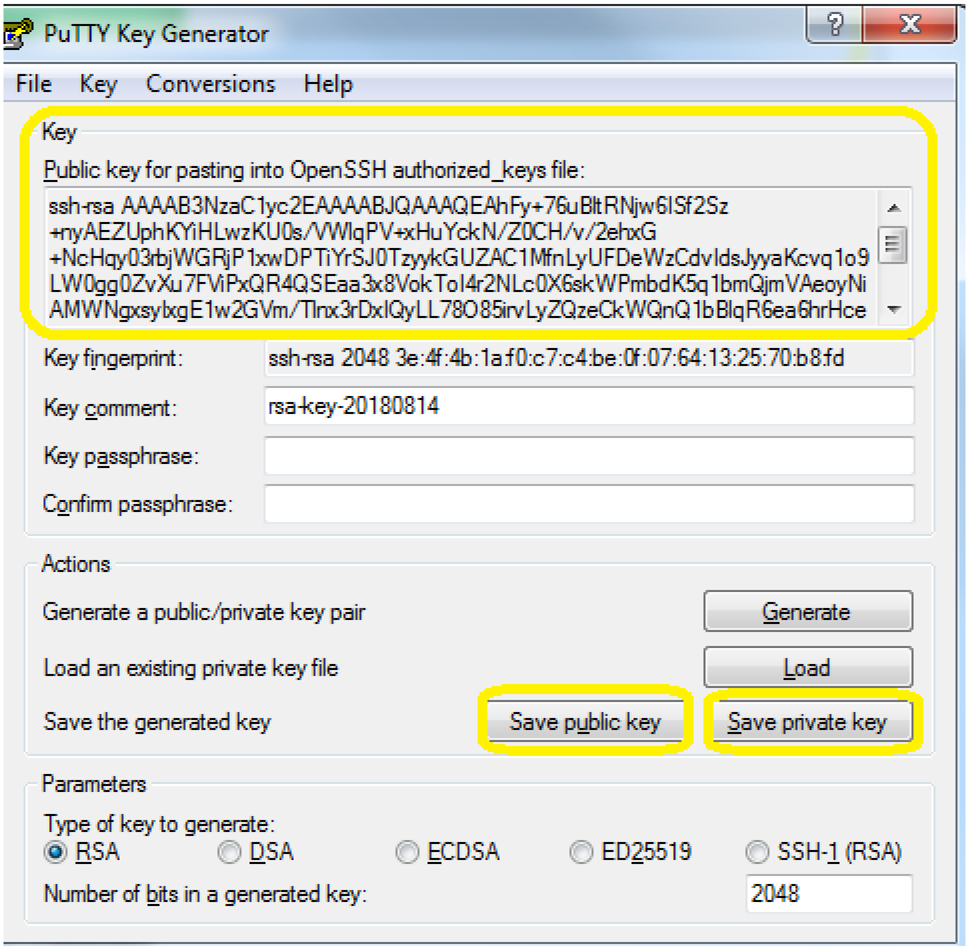




The default settings are fine; we will generate a 2048-bit RSA keypair, Hit the generate button. To generate a random key, PuTTY key generator uses the input of your mouse movement. Swing the mouse around a bit until the keys are ready.



Hit the save public key and save private key buttons to save the public and private Keys.

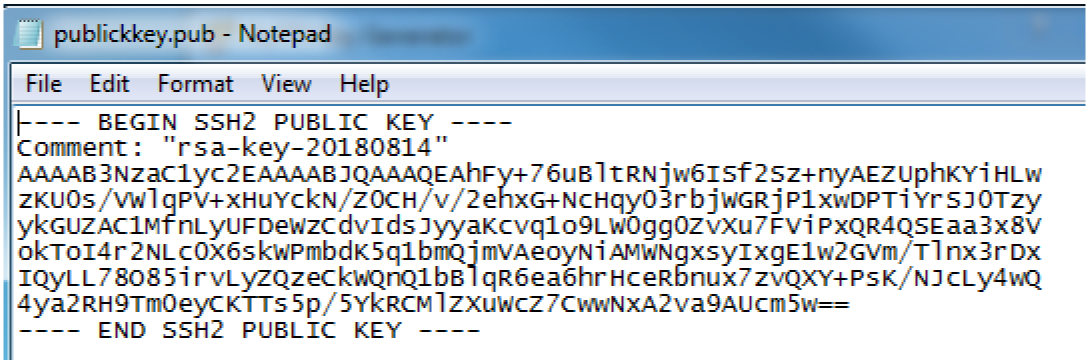


Cisco Router IOS Configuration:

Let us start with a basic SSH configuration. First, configure hostname & domain name:

R1(config)#ip domain-name test.com
R1(config)#crypto key generate rsa modulus 2048
R1(config)#ip ssh version 2
R1(config)#line vty 0 4
R1(config-line)#transport input ssh
R1(config-line)#login local

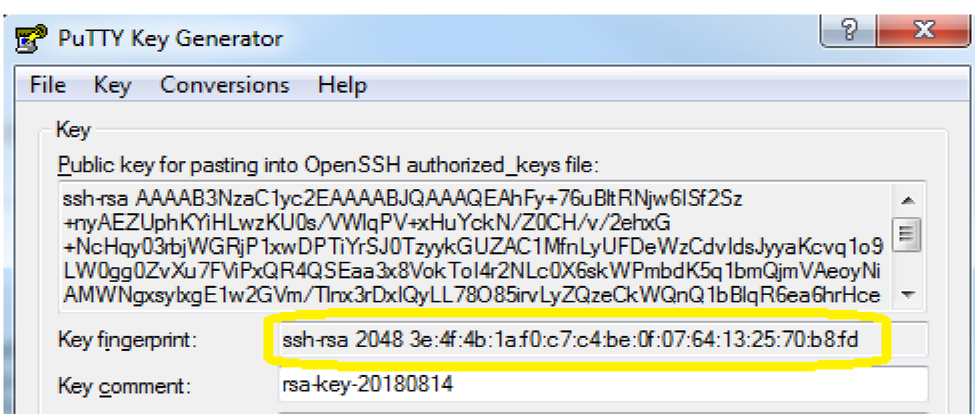
Open the public key file (publickey.pub) in your favorite text editor.



Add the public key for a username we choose in this case I will call this user "admin".

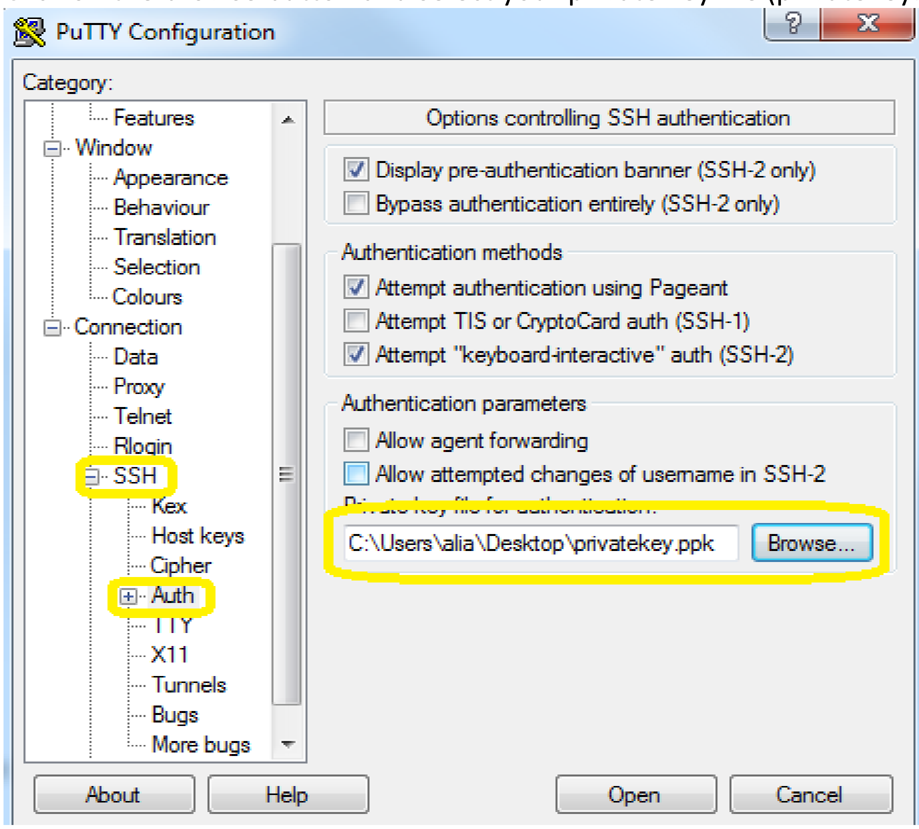
Once you enter key-string command, you can keep adding lines until you type exit:

R1(config)#ip ssh pubkey-chain
R1(conf-ssh-pubkey)#username admin
R1(conf-ssh-pubkey-user)#key-string
R1(conf-ssh-pubkey-data)# AAAAB3NzaC1yc2EAAAABJQAAAQEAhFy+76uBltRNjw6ISf2Sz+nyAEZUphKYiHLw zKU0s/VWlqPV+xHuYckN/Z0CH/v/2ehxG+NcHqy03rbjWGRjP1xwDPTiYrSJ0Tzy R1(conf-ssh-pubkey-data)# ykGUZAC1MfnLyUFDeWzCdvIdsJyyaKcvq1o9LW0gg0ZvXu7FViPxQR4QSEaa3x8V okToI4r2NLc0X6skWPmbdK5q1bmQjmVAeoyNiAMWNgxsyIxgE1w2GVm/Tlnx3rDx R1(conf-ssh-pubkey-data)# IQyLL78O85irvLyZQzeCkWQnQ1bBlqR6ea6hrHceRbnux7zvQXY+PsK/NJcLy4wQ 4ya2RH9Tm0eyCKTTs5p/5YkRCMlZXuWcZ7CwwNxA2va9AUcm5w== R1(conf-ssh-pubkey-data)#exit
Our router now knows the public key of the users. R1#show running-config | begin pubkey

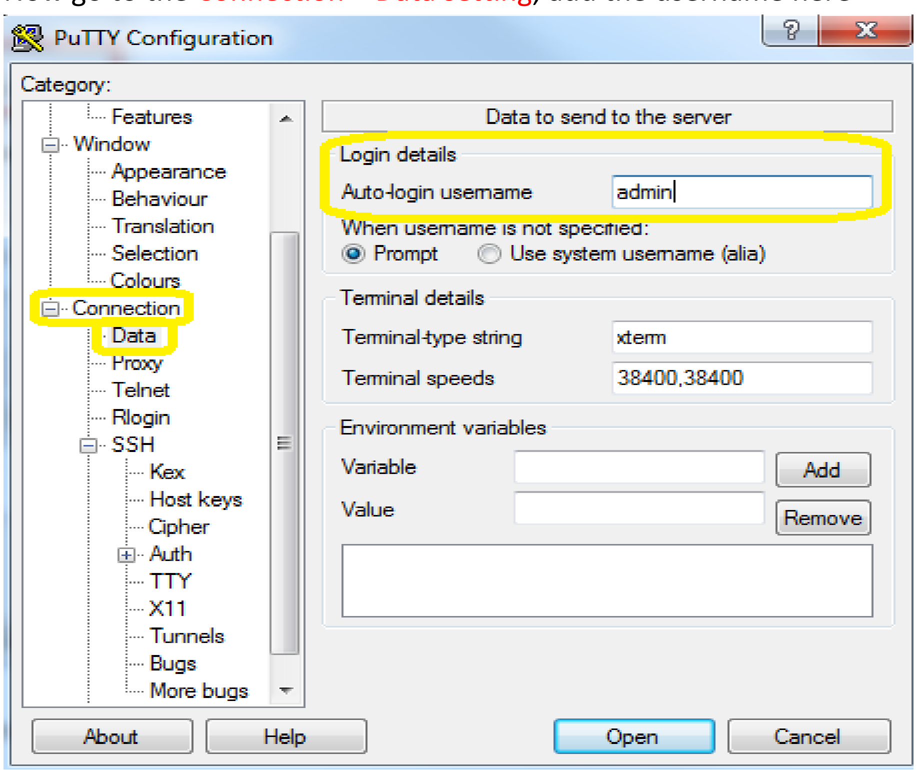


Time to configure PuTTY. Open PuTTY and look for the Connection > SSH setting.

Click on the browse button and select your private key file (privatekey.ppk):



Now go to the Connection > Data setting, add the username here



Go to the main screen and if you do not want to lose these settings, save your session.